Written HW 8 — Full Solutions

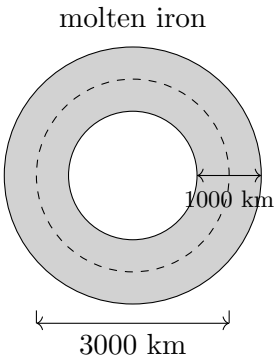
Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 29. $\mu_0 = 4\pi \times 10^{-7}$ T · m/A throughout.

29.50 (3 pts)

The earth's magnetic dipole moment, 8.0×10^{22} A · m^2, is modeled as arising from current flowing around a 3000-km-diameter loop made from a very fat, 1000-km-diameter "wire" of molten iron (both diameters measured from the wire's centerline). (a) What current flows in the loop? (b) What is the current density J in the loop? (c) Compare that to the current density of an ordinary 1.0 A current in a 1.0-mm-diameter wire — is the earth's current density large or small?

MODEL: A current loop's dipole moment is $\mu = IA_{\text{loop}}$, where A_{loop} is the area enclosed by the loop's *centerline* — that's what fixes the current, regardless of how fat the wire is. Current *density* is a separate idea: $J = I/A_{\text{wire}}$, the same current spread over the wire's own cross-sectional area. Here those two areas are wildly different, since the "wire" is nearly as big as the loop itself.

VISUALIZE:



The dashed circle is the loop's centerline (radius 1500 km, fixing A_{loop}); the gray annulus is the wire's own cross section (radius 500 km, fixing A_{wire}).

SOLVE: (a)

$$A_{\text{loop}} = \pi r_{\text{loop}}^2 = \pi(1500 \times 10^3 \text{ m})^2 = 7.07 \times 10^{12} \text{ m}^2$$

$$I = \frac{\mu}{A_{\text{loop}}} = \frac{8.0 \times 10^{22} \text{ A} \cdot \text{m}^2}{7.07 \times 10^{12} \text{ m}^2} = \boxed{1.13 \times 10^{10} \text{ A}}.$$

(b)

$$A_{\text{wire}} = \pi r_{\text{wire}}^2 = \pi(500 \times 10^3 \text{ m})^2 = 7.85 \times 10^{11} \text{ m}^2$$

$$J_{\text{loop}} = \frac{I}{A_{\text{wire}}} = \frac{1.13 \times 10^{10} \text{ A}}{7.85 \times 10^{11} \text{ m}^2} = \boxed{0.014 \text{ A/m}^2}.$$

(c)

$$J_{\text{wire}} = \frac{I}{A} = \frac{1.0 \text{ A}}{\pi(0.50 \times 10^{-3} \text{ m})^2} = \boxed{1.3 \times 10^6 \text{ A/m}^2}.$$

REVIEW: $J_{\text{loop}} \ll J_{\text{wire}}$ by about eight orders of magnitude. The earth's core really does carry a staggering total current (1.13×10^{10} A, tens of millions of times a lightning bolt), but spread over a

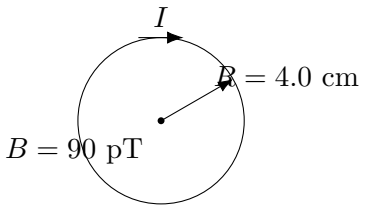
cross section 500 km across, the current *density* — how crowded the charge carriers actually are — is tiny. Total current and current density answer two different questions, and this problem is a good reminder that a huge I doesn't automatically mean a huge J.

29.52 (3 pts)

The heart produces a weak magnetic dipole field from current circulating in its outer layers, modeled as a current loop. (a) If the field at the center of an 8.0-cm-diameter loop (about the size of a human heart) is 90 pT, what current circulates around it? (b) What is the magnitude of the heart's magnetic dipole moment?

MODEL: A current loop's field at its own center is $B = \mu_0 I/(2R)$; the same loop's dipole moment is $\mu = IA = I\pi R^2$. Solve the first relation for I, then use that current to get μ.

VISUALIZE:



The loop's radius is half the given 8.0 cm diameter, $R = 4.0$ cm.

SOLVE: (a)

$$B = \frac{\mu_0 I}{2R} \Rightarrow I = \frac{2BR}{\mu_0} = \frac{2(90 \times 10^{-12}\ \text{T})(0.040\ \text{m})}{4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}} = \boxed{5.7 \times 10^{-6}\ \text{A} = 5.7\ \mu\text{A}}.$$

(b)

$$\mu = I\pi R^2 = (5.73 \times 10^{-6}\ \text{A})\,\pi(0.040\ \text{m})^2 = \boxed{2.9 \times 10^{-8}\ \text{A}\cdot\text{m}^2}.$$

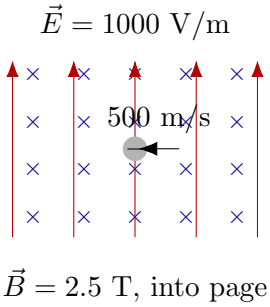
REVIEW: Side by side with 29.50, the heart's dipole moment ($2.9 \times 10^{-8}\ \text{A}\cdot\text{m}^2$) is about 10^{30} times smaller than the earth's ($8.0 \times 10^{22}\ \text{A}\cdot\text{m}^2$) — a vivid sense of just how faint biomagnetic fields are, and why detecting them (magnetocardiography) takes extraordinarily sensitive equipment.

29.61 (4 pts)

An antiproton (same mass as a proton, but $q = -e$) moves at 500 m/s to the left through a region with uniform fields $\vec{E} = 1000$ V/m (pointing up, in the plane of the page) and $\vec{B} = 2.5$ T (into the page). What are the magnitude and direction of the antiproton's acceleration at this instant?

MODEL: The electric and magnetic forces act simultaneously; the net force is just their vector sum, $\vec{F}_{\text{net}} = q\vec{E} + q\vec{v} \times \vec{B}$, and then $\vec{a} = \vec{F}_{\text{net}}/m$. Nothing here requires the two forces to cancel — that's a different (and more special) situation, the one we'll meet in 29.59.

VISUALIZE:



The antiproton is negative, so its electric force is *opposite* $\vec{E}$ (down), while its magnetic force, by the right-hand rule applied to a negative charge moving left through a field into the page, works out to point *up*.

SOLVE:

$$F_E = eE = (1.60 \times 10^{-19}\text{ C})(1000\text{ V/m}) = 1.60 \times 10^{-16}\text{ N, down}$$

$$F_B = evB = (1.60 \times 10^{-19}\text{ C})(500\text{ m/s})(2.5\text{ T}) = 2.00 \times 10^{-16}\text{ N, up}$$

$$F_{\text{net}} = F_B - F_E = 0.40 \times 10^{-16}\text{ N, up} = 4.0 \times 10^{-17}\text{ N, up}$$

$$a = \frac{F_{\text{net}}}{m} = \frac{4.0 \times 10^{-17}\text{ N}}{1.67 \times 10^{-27}\text{ kg}} = \boxed{2.4 \times 10^{10}\text{ m/s}^2\text{, up}}.$$

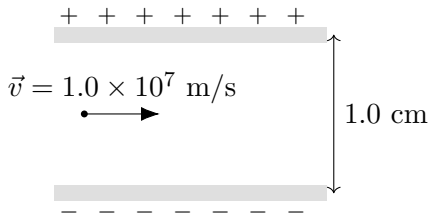
REVIEW: The two forces are close in size (within 25% of each other) but don't cancel at this speed, so the antiproton still accelerates hard — with a proton-scale mass, even a 10^{-17} N force gives a 10^{10} m/s^2 acceleration. 29.59 asks the reverse question: given a speed, what field makes these two forces *exactly* cancel? Same two force expressions, just solved for a different unknown.

29.59 (4 pts)

An electron travels at 1.0×10^7 m/s between two parallel plates separated by 1.0 cm and charged by a 200 V battery (top plate positive). What magnetic field strength and direction will let the electron pass between the plates without being deflected?

MODEL: Same two force expressions as 29.61, $F_{\text{elec}} = qE$ and $F_{\text{mag}} = qvB$ — but now we want them to exactly cancel, so the electron feels zero net force and travels straight through. That special condition, $F_{\text{elec}} = F_{\text{mag}}$, is the velocity-selector balance.

VISUALIZE:



With the top plate positive, $\vec{E}$ points down (toward the negative plate). The electron, being negative, feels an electric force *up* (opposite $\vec{E}$), so the magnetic force needs to point *down* to cancel it.

SOLVE:

$$E = \frac{\Delta V}{d} = \frac{200 \text{ V}}{0.010 \text{ m}} = 2.0 \times 10^4 \text{ V/m}$$

$$F_{\text{elec}} = eE = (1.60 \times 10^{-19} \text{ C})(2.0 \times 10^4 \text{ V/m}) = 3.2 \times 10^{-15} \text{ N, up}$$

Setting $F_{\text{mag}} = F_{\text{elec}}$ (down, to cancel the upward electric force):

$$evB = F_{\text{elec}} \;\Rightarrow\; B = \frac{F_{\text{elec}}}{ev} = \frac{3.2 \times 10^{-15} \text{ N}}{(1.60 \times 10^{-19} \text{ C})(1.0 \times 10^7 \text{ m/s})} = 2.0 \times 10^{-3} \text{ T}.$$

For an electron moving right to feel a magnetic force pointing down, the right-hand rule (flipped for the negative charge) requires $\vec{B}$ into the page:

$$\vec{B} = \boxed{2.0 \text{ mT, into the page}}.$$

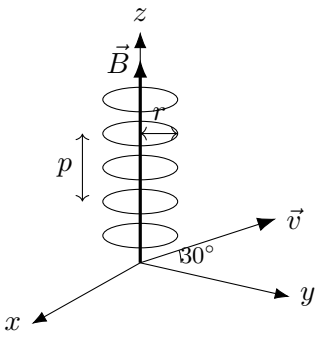
REVIEW: Check: $E/B = (2.0 \times 10^4 \text{ V/m})/(2.0 \times 10^{-3} \text{ T}) = 1.0 \times 10^7$ m/s — exactly the given electron speed, confirming the balance condition $v = E/B$ (the same relation Prep 24's velocity selector uses). Compare to 29.61: there, the fields were given numbers that did *not* satisfy $v = E/B$, so the particle accelerated instead of passing straight through.

29.65 (4 pts)

A uniform 30 mT magnetic field points in the $+z$ direction. An electron enters the field moving at 5.0×10^6 m/s, at 30° above the xy-plane. Find the radius r and the pitch p of the electron's spiral (helical) trajectory.

MODEL: Split $\vec{v}$ into a component parallel to $\vec{B}$ and a component perpendicular to it. The parallel component feels no magnetic force (since $\vec{v} \times \vec{B} = 0$ when they're parallel) and drifts at constant speed along $\vec{B}$; the perpendicular component feels the usual magnetic force and drives circular motion. Together they trace out a helix: r comes from the perpendicular speed, and the pitch p (distance advanced per full revolution) comes from the parallel speed times the period of that circular motion.

VISUALIZE:



The 30° is measured from the xy-plane (not from $\vec{B}$), so $v_\parallel = v \sin 30°$ and $v_\perp = v \cos 30°$.

SOLVE:

$$v_\perp = v \cos 30° = (5.0 \times 10^6 \text{ m/s})(0.866) = 4.33 \times 10^6 \text{ m/s}$$

$$v_\parallel = v \sin 30° = (5.0 \times 10^6 \text{ m/s})(0.500) = 2.50 \times 10^6 \text{ m/s}$$

$$r = \frac{mv_\perp}{eB} = \frac{(9.11 \times 10^{-31} \text{ kg})(4.33 \times 10^6 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(0.030 \text{ T})} = \boxed{0.82 \text{ mm}}$$

$$T = \frac{2\pi r}{v_\perp} = \frac{2\pi(8.2 \times 10^{-4} \text{ m})}{4.33 \times 10^6 \text{ m/s}} = 1.19 \times 10^{-9} \text{ s}$$

$$p = v_\parallel T = (2.50 \times 10^6 \text{ m/s})(1.19 \times 10^{-9} \text{ s}) = \boxed{3.0 \text{ mm}}.$$

REVIEW: As a check, the period doesn't actually need $v_\perp$ at all — $T = 2\pi m/(eB) = 2\pi(9.11 \times 10^{-31})/[(1.60 \times 10^{-19})(0.030)] = 1.19 \times 10^{-9}$ s, the same value, since the cyclotron period is independent of speed. Most of the electron's speed goes into the perpendicular, circular part of the motion ($\cos 30° \approx 0.87$) and less into the parallel drift ($\sin 30° = 0.50$) — yet the pitch (3.0 mm) still comes out almost four times the radius (0.82 mm), since the pitch accumulates over a *full period*, not just a comparison of speed components.